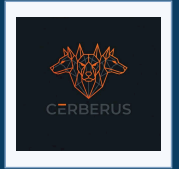


Leverage Flow Regime



INTERNAL

Combines Binance hourly taker buy/sell flow, long/short positioning and the fear-and-greed index into a leverage-flow crowding regime.

Author: Cerberus Research

Generated: 2026-08-05 08:52 UTC

N

38

EDGE

-3.01%

P

0.00832

HIT-RATE

34.2%

CLASSIFICATION

internal

AUTHOR

Cerberus Research

RUN ID

20260805_095243

GIT COMMIT

ec177e3b

DURATION

1.1s

DATASETS

duckdb:crypto_binance_taker_ls_ratio, duckdb:crypto_binance_global_ls_ratio, duckdb:crypto_binance_top_ls_positions, duckdb:crypto_fear_greed_index, duckdb:crypto_market_data

PARAMETER · SYMBOL

BTCUSDT

PARAMETER · BENCHMARK

BTC-USD

PARAMETER · LOOKBACK

30

PARAMETER · HORIZON

7

PARAMETER · Z_THRESHOLD

2

PARAMETER · MIN_OBS

200

PARAMETER · MIN_DAILY_OBS

60

KPI

Value

N

38

EDGE

-3.01%

P

0.00832

HIT-RATE

34.2%

KEY FINDINGS

- Latest BTCUSDT flow z: -0.27 | net buy ratio: +0.42% | corr with BTC 7d forward: -0.16
- Top flow quintile (Q5): BTC avg 7d fwd return -3.01% (n=38, hit-rate 34.2%)
- Latest fear-and-greed: 27 (Fear)

CONTENTS

- 01 Executive Summary
- 02 Results & Interpretation
- 03 Key Findings
- 04 Data Sources
- 05 Objectives
- 06 Methodology
- 07 Charts
- 08 Statistical Results
- 09 Limitations

Generated by the Cerberus studies framework · schema 2

Data Sources

- **duckdb:crypto_binance_taker_ls_ratio**
- **duckdb:crypto_binance_global_ls_ratio**
- **duckdb:crypto_binance_top_ls_positions**
- **duckdb:crypto_fear_greed_index**
- **duckdb:crypto_market_data**

Key Findings

- Latest BTCUSDT flow z: -0.27 | net buy ratio: +0.42% | corr with BTC 7d forward: -0.16
- Top flow quintile (Q5): BTC avg 7d fwd return -3.01% (n=38, hit-rate 34.2%)
- Latest fear-and-greed: 27 (Fear)
- Dropped 17 symbol(s) with insufficient coverage: ALGOUSDT, WIFUSDC, SUSHIUSDT, EGLDUSDT, JUPUSDT, UNIUSDT, TIAUSDT, LDOUSDT, WIFUSDT, FILUSDT, FETUSDT, XRPUSDT, ETCUSDT, ARBUSDT, DOTUSDT, DOGEUSDT, AVAXUSDT

Executive Summary

NOTE

Combines Binance hourly taker buy/sell flow, long/short positioning and the fear-and-greed index into a leverage-flow crowding regime and measures its BTC forward-return impact.

Results & Interpretation

- Across 38 events, the forward return averaged -3.01%. Returns were positive 34.2% of the time. The difference has $p \approx 0.00832$.
- Across 37 events, the forward return averaged +0.03%. Returns were positive 59.5% of the time. The difference has $p \approx 0.98$.
- Across 37 events, the forward return averaged -1.39%. Returns were positive 40.5% of the time. The difference has $p \approx 0.139$.
- Across 37 events, the forward return averaged +0.03%. Returns were positive 59.5% of the time. The difference has $p \approx 0.972$.
- Across 38 events, the forward return averaged -0.74%. Returns were positive 36.8% of the time. The difference has $p \approx 0.314$.

Objectives

NOTE

Evaluate whether the studied signal produces a measurable, repeatable edge.

Quantify the reliability of the observed patterns across the sample.

Methodology

Hourly taker buy/sell volume and long/short ratios are aggregated daily for BTCUSDT, z-scored over a 30-day window and averaged into a flow composite. Flow spikes fire when $|z| > 2.0$; BTC 7d forward returns are bucketed by flow quintile with t-tests. The fear-and-greed index is overlaid as a sentiment context.

Charts

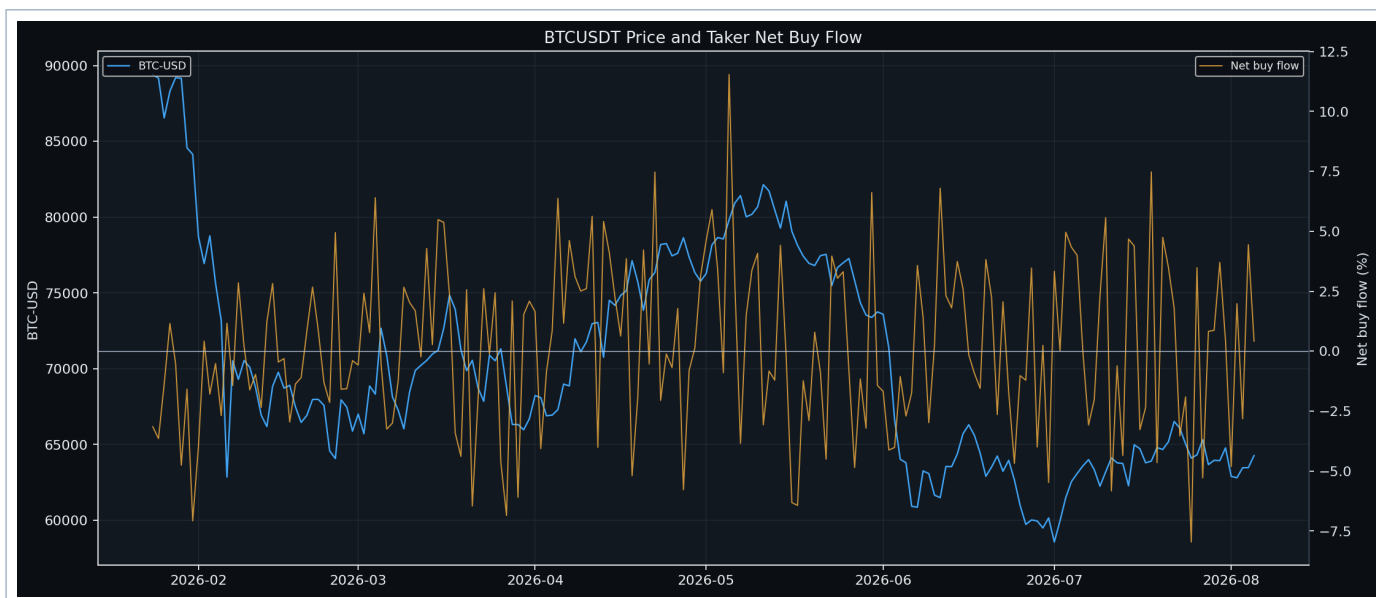


Figure 1. BTCUSDT BTC-USD price, net buy-flow ratio and flow z
BTCUSDT BTC-USD price, net buy-flow ratio and flow z

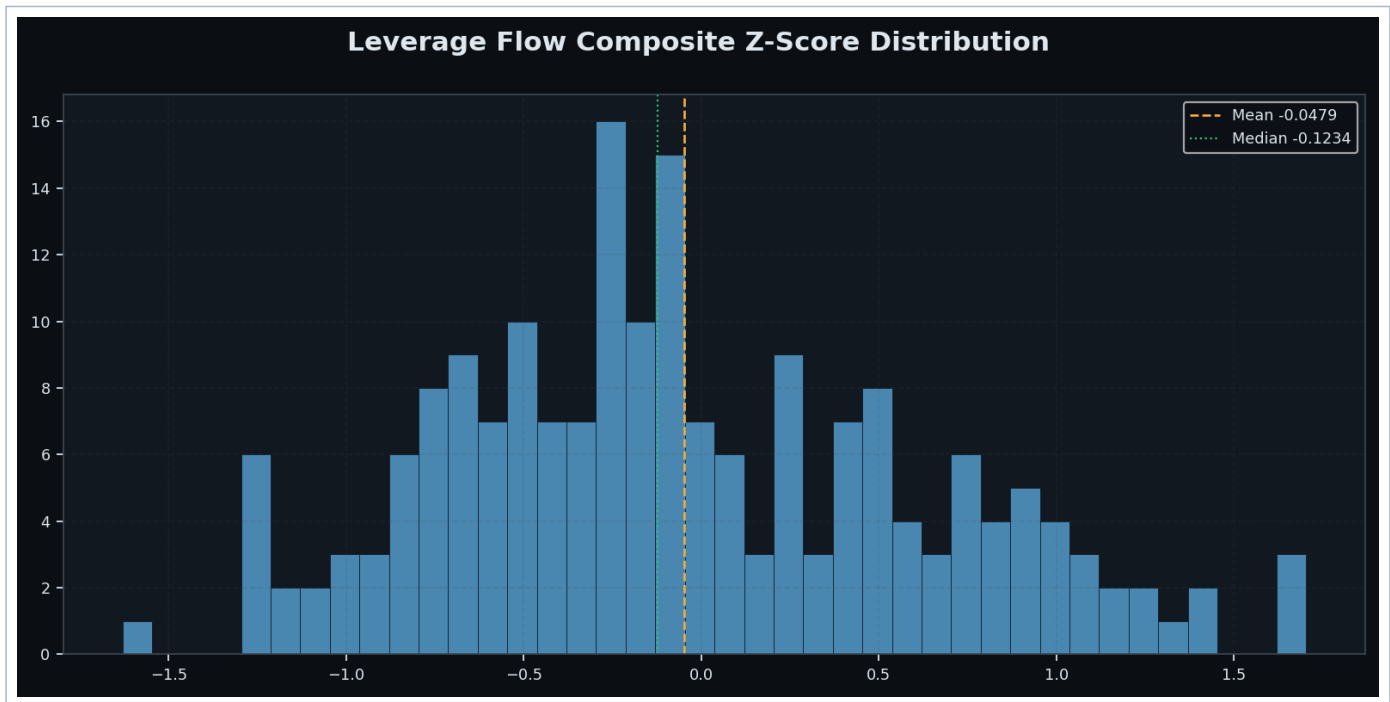


Figure 2. Distribution of 30-day flow z-scores

Distribution of 30-day flow z-scores_

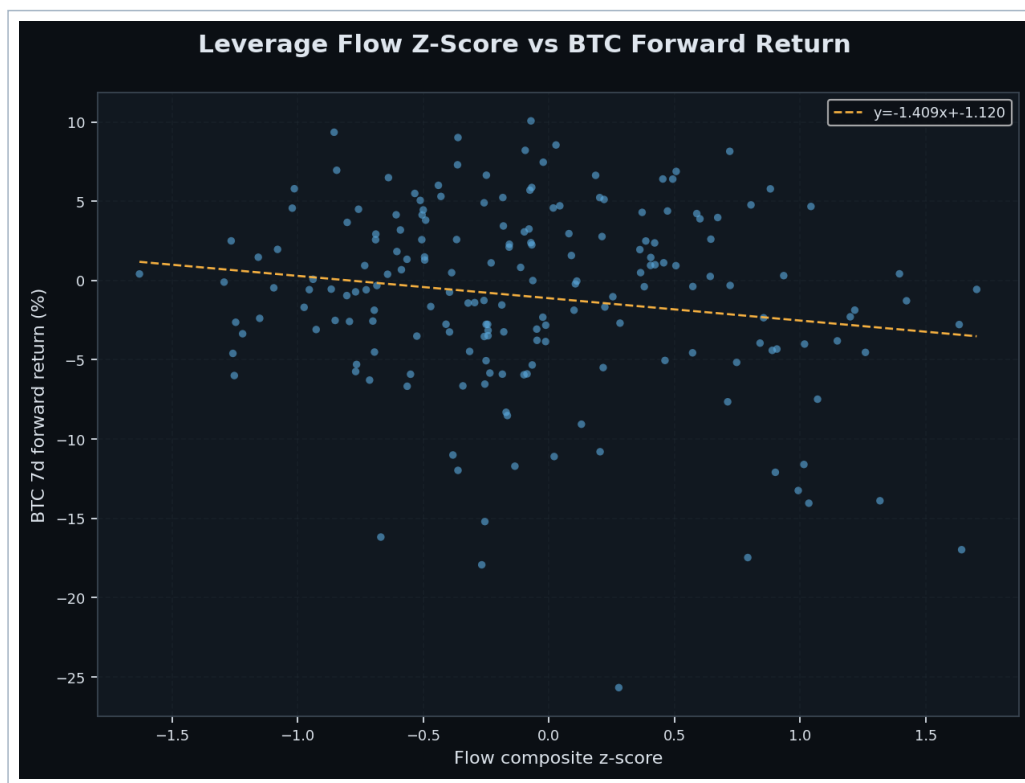


Figure 3. BTC 7d forward return vs leverage flow z

BTC 7d forward return vs leverage flow z_

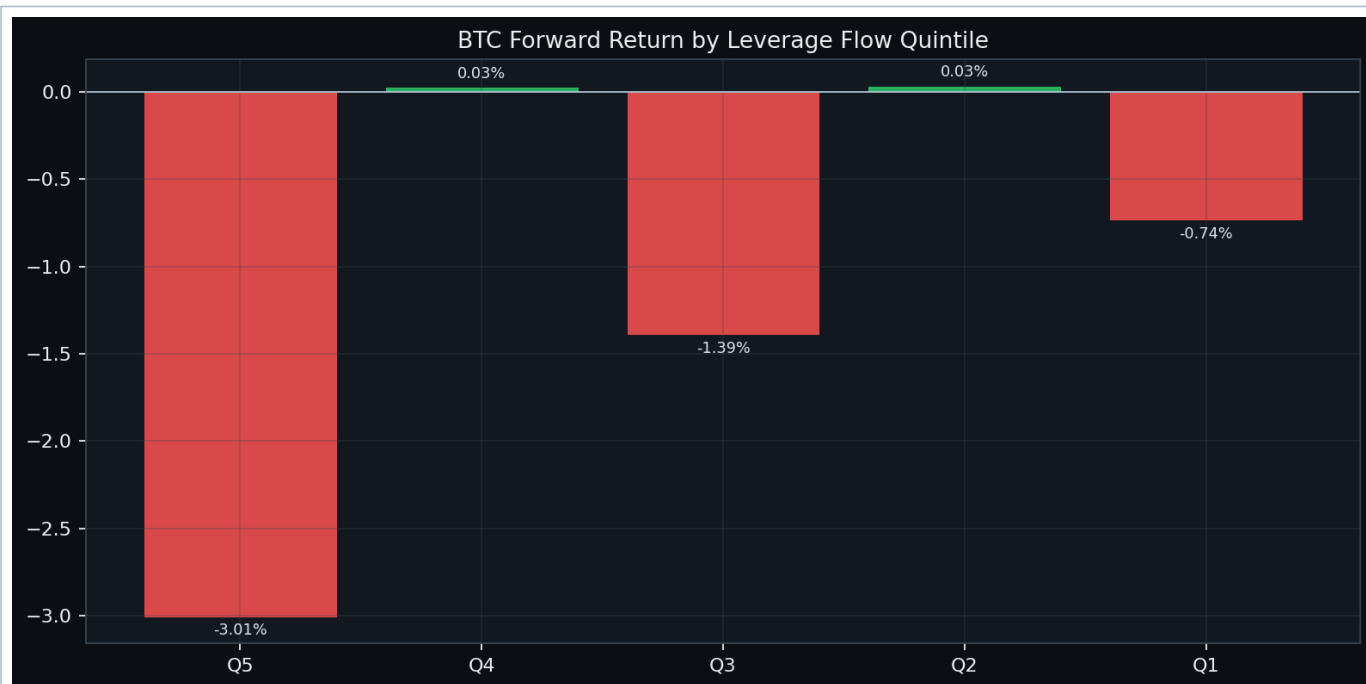


Figure 4. BTC 7d forward return by flow quintile

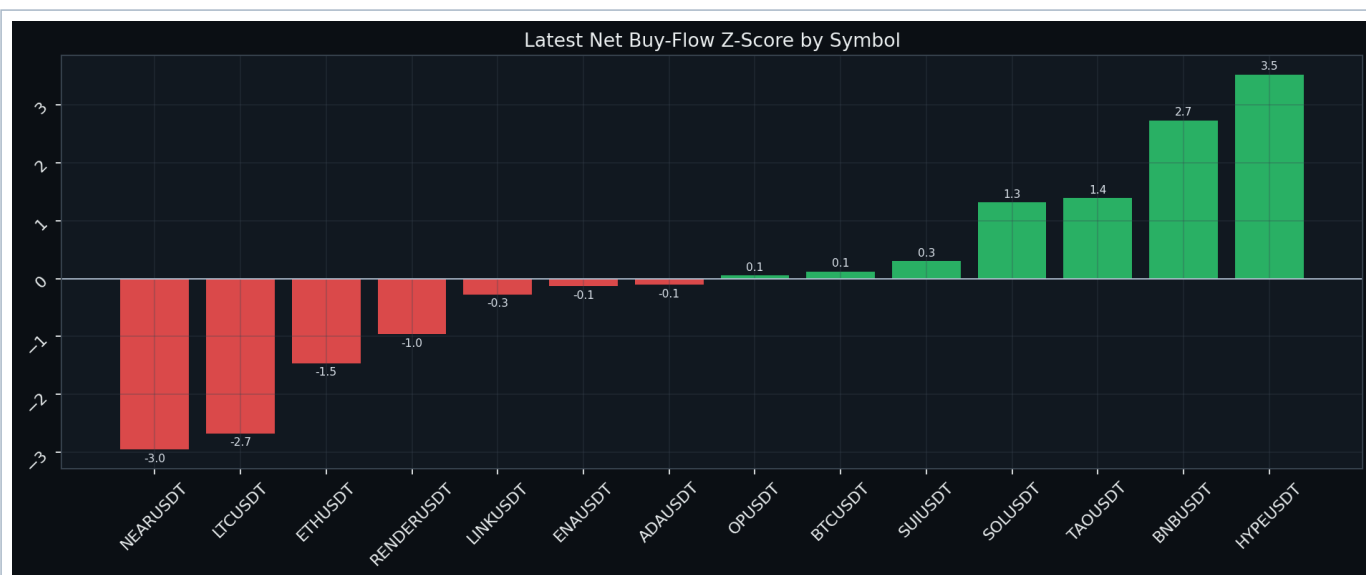
BTC 7d forward return by flow quintile

Figure 5. Latest net buy-flow z-score by symbol

Latest net buy-flow z-score by symbol

Statistical Results

close	net_buy_ratio_...	ls_ratio	flow_z	fear_greed	btc_fwd_ret_pct
60,861.88	+ 3.58%	2.01	0.88	12	+ 5.78%
63,254.57	+ 1.41%	1.89	0.60	8	+ 3.89%
63,078.44	-2.98%	2.00	0.22	10	+ 5.11%
61,658.31	+ 0.25%	2.06	0.49	9	+ 6.39%
61,492.51	+ 6.80%	1.75	0.80	12	+ 4.76%

close	net_buy_ratio_...	ls_ratio	flow_z	fear_greed	btc_fwd_ret_pct
63,552.30	+ 2.32%	1.55	0.26	12	-1.03%
63,537.74	+ 1.81%	1.50	0.11	13	-0.04%
64,377.58	+ 3.74%	1.43	0.11	18	-0.21%
65,713.62	+ 2.60%	1.35	-0.05	20	-3.78%
66,300.96	-0.15%	1.39	-0.26	23	-3.54%
65,598.94	-0.93%	1.53	-0.31	22	-4.48%
64,422.57	-1.55%	1.83	-0.07	15	-5.33%
62,900.23	+ 3.82%	1.89	0.46	14	-5.05%
63,513.66	+ 2.25%	1.78	0.22	23	-5.50%
64,240.23	-2.64%	1.61	-0.56	23	-6.67%
63,231.87	+ 2.06%	1.60	-0.18	20	-5.91%
63,957.20	-1.70%	1.97	-0.10	23	-5.95%
62,662.29	-4.68%	2.22	-0.25	17	-6.54%
60,989.57	-1.02%	2.31	0.22	12	-1.68%
59,726.74	-1.21%	2.13	0.08	13	+ 2.95%
60,022.33	+ 3.47%	2.03	0.59	15	+ 4.22%
59,953.44	-3.99%	2.08	-0.18	18	+ 5.23%
59,494.61	+ 0.24%	2.20	0.51	12	+ 6.88%
60,152.36	-5.47%	2.60	0.45	15	+ 6.40%
58,566.09	+ 3.33%	2.24	0.72	11	+ 8.14%
59,967.69	+ 0.01%	1.78	-0.49	19	+ 3.80%
61,490.21	+ 4.96%	1.69	0.21	21	+ 2.77%
62,553.61	+ 4.33%	1.51	0.39	22	+ 2.49%
63,088.41	+ 4.00%	1.46	0.46	23	+ 1.11%
63,586.10	-0.01%	1.47	0.64	24	+ 0.25%
64,001.24	-3.08%	1.48	0.28	27	-2.69%
63,335.31	-2.02%	1.70	0.65	20	+ 2.60%
62,247.13	+ 2.31%	1.59	0.67	22	+ 3.97%
63,192.88	+ 5.57%	1.30	0.41	23	+ 0.95%
64,110.97	-5.83%	1.26	-0.68	26	-0.32%
63,791.35	-0.61%	1.31	0.09	26	+ 1.57%
63,746.66	-4.35%	1.60	0.41	28	+ 1.45%

Research only. Not investment advice. Past performance is not indicative of future results.

close	net_buy_ratio_...	ls_ratio	flow_z	fear_greed	btc_fwd_ret_pct
62,278.96	+ 4.68%	1.55	1.04	22	+ 4.67%
64,983.52	+ 4.39%	1.19	0.42	25	+ 2.37%
64,717.17	-3.27%	1.31	-0.16	25	+ 2.10%
63,792.29	-2.34%	1.67	0.36	27	+ 1.94%
63,904.95	+ 7.47%	1.55	0.94	25	+ 0.30%
64,795.70	-4.65%	1.37	-0.40	28	-0.74%
64,670.40	+ 4.75%	1.34	0.42	29	+ 0.99%
65,184.92	+ 3.49%	1.10	-0.02	25	-2.32%
66,520.98	+ 1.79%	1.12	-0.01	33	-3.85%
66,077.06	-3.53%	1.37	-0.18	31	-3.24%
65,033.02	-1.90%	1.67	0.38	28	-0.39%
64,099	-7.96%	1.84	0.10	27	-1.88%
64,316.36	+ 3.48%	1.75	0.86	26	-2.35%
65,310.39	-5.28%	1.55	-0.01	30	-2.82%
63,673.71	+ 0.83%	1.81	0.72	29	-0.32%
63,957.04	+ 0.87%	1.56	0.37	29	+ 0.49%
63,934.10	+ 3.70%	1.39	0.31	28	N/A
64,776.94	+ 0.44%	1.80	0.73	25	N/A
62,896.51	-4.81%	2.15	0.76	27	N/A
62,802.63	+ 1.98%	1.99	1.06	27	N/A
63,466.47	-2.82%	1.85	0.47	28	N/A
63,472.83	+ 4.44%	1.42	0.27	25	N/A
64,271.53	+ 0.42%	1.27	-0.27	27	N/A

Table 1. Daily leverage flow frame tail

time	event	net_buy_z	net_buy_ratio_...	close
2026-01-31	SELL FLOW SPIKE	-2.08	-7.07%	84,141.78
2026-02-25	BUY FLOW SPIKE	2.30	+ 4.95%	64,074.11
2026-03-04	BUY FLOW SPIKE	2.83	+ 6.40%	68,321.62
2026-03-21	SELL FLOW SPIKE	-2.24	-6.45%	70,552.63
2026-03-27	SELL FLOW SPIKE	-2.02	-6.85%	68,791.11
2026-05-05	BUY FLOW SPIKE	2.56	+ 11.54%	79,823.89
2026-06-11	BUY FLOW SPIKE	2.12	+ 6.80%	61,492.51

Table 2. Detected buy/sell flow spike events

regime	count	avg_fwd_ret_...	median_fwd_ret_...	hit_rate_...	t_stat	p_value
Q5	38	-3.01%	-2.32%	+ 34.21%	-2.79	+ 0.00832
Q4	37	+ 0.03%	+ 1.11%	+ 59.46%	0.03	- 0.98
Q3	37	-1.39%	-2.77%	+ 40.54%	-1.51	- 0.139
Q2	37	+ 0.03%	+ 1.29%	+ 59.46%	0.04	- 0.972
Q1	38	-0.74%	-0.58%	+ 36.84%	-1.02	- 0.314

Table 3. BTC forward returns by flow quintile

symbol	hourly_ro...	daily_rows	latest_net_buy_...	latest_flo...	retained
ALGOUSDT	485	21	N/A	N/A	False
WIFUSDC	485	21	N/A	N/A	False
SUSHIUSDT	485	21	N/A	N/A	False
EGLDUSDT	485	21	N/A	N/A	False
JUPUSDT	653	28	N/A	N/A	False
UNIUSDT	653	28	N/A	N/A	False
TIAUSDT	653	28	N/A	N/A	False
LDOUSDT	653	28	N/A	N/A	False
WIFUSDT	653	28	N/A	N/A	False
FILUSDT	653	28	N/A	N/A	False
FETUSDT	653	28	N/A	N/A	False
XRPUSDT	653	28	N/A	N/A	False
ETCUSDT	653	28	N/A	N/A	False
ARBUSDT	653	28	N/A	N/A	False
DOTUSDT	653	28	N/A	N/A	False
DOGEUSDT	653	28	N/A	N/A	False
AVAXUSDT	653	28	N/A	N/A	False
HYPEUSDT	5056	212	+ 12.71%	3.53	True
BNBUSDT	5314	223	+ 7.56%	2.73	True
TAOUSDT	5314	223	+ 1.06%	1.39	True
BTCUSDT	5314	223	+ 0.42%	0.12	True
SUIUSDT	5314	223	-1.88%	0.30	True
RENDERUSDT	5314	223	-8.15%	-0.96	True
ETHUSDT	5314	223	-5.49%	-1.46	True

symbol	hourly_ro...	daily_rows	latest_net_buy_...	latest_flo...	retained
OPUSDT	5314	223	-1.89%	0.05	True
NEARUSDT	5314	223	-18.45%	-2.96	True
LTCUSDT	5314	223	-11.04%	-2.67	True
LINKUSDT	5314	223	-1.96%	-0.28	True
ENAUUSD	5314	223	-2.66%	-0.13	True
SOLUSDT	5314	223	+ 3.35%	1.32	True
ADAUSDT	5314	223	-1.05%	-0.11	True

Table 4. Symbol coverage and latest flow z-scores

Limitations

NOTE

Results are research-grade, not investment advice. The sample is limited to available data, and measured edges may not persist out-of-sample.

Appendix Tables

flow_events

time	event	net_buy_z	net_buy_ratio_...	close
2026-01-31	SELL FLOW SPIKE	-2.08	-7.07%	84,141.78
2026-02-25	BUY FLOW SPIKE	2.30	+ 4.95%	64,074.11
2026-03-04	BUY FLOW SPIKE	2.83	+ 6.40%	68,321.62
2026-03-21	SELL FLOW SPIKE	-2.24	-6.45%	70,552.63
2026-03-27	SELL FLOW SPIKE	-2.02	-6.85%	68,791.11
2026-05-05	BUY FLOW SPIKE	2.56	+ 11.54%	79,823.89
2026-06-11	BUY FLOW SPIKE	2.12	+ 6.80%	61,492.51

frame

date	close	net_buy_ratio_...	ls_ratio	flow_z	fear_greed	btc_fwd_ret_pct
2026-06-07	60,861.88	+ 3.58%	2.01	0.88	12	+ 5.78%
2026-06-08	63,254.57	+ 1.41%	1.89	0.60	8	+ 3.89%
2026-06-09	63,078.44	-2.98%	2.00	0.22	10	+ 5.11%
2026-06-10	61,658.31	+ 0.25%	2.06	0.49	9	+ 6.39%
2026-06-11	61,492.51	+ 6.80%	1.75	0.80	12	+ 4.76%
2026-06-12	63,552.30	+ 2.32%	1.55	0.26	12	-1.03%

date	close	net_buy_ratio_...	ls_ratio	flow_z	fear_greed	btc_fwd_ret_pct
2026-06-13	63,537.74	+ 1.81%	1.50	0.11	13	-0.04%
2026-06-14	64,377.58	+ 3.74%	1.43	0.11	18	-0.21%
2026-06-15	65,713.62	+ 2.60%	1.35	-0.05	20	-3.78%
2026-06-16	66,300.96	-0.15%	1.39	-0.26	23	-3.54%
2026-06-17	65,598.94	-0.93%	1.53	-0.31	22	-4.48%
2026-06-18	64,422.57	-1.55%	1.83	-0.07	15	-5.33%
2026-06-19	62,900.23	+ 3.82%	1.89	0.46	14	-5.05%
2026-06-20	63,513.66	+ 2.25%	1.78	0.22	23	-5.50%
2026-06-21	64,240.23	-2.64%	1.61	-0.56	23	-6.67%

regime_forward_returns

regime	count	avg_fwd_ret_...	median_fwd_ret_...	hit_rate_...	t_stat	p_value
Q5	38	-3.01%	-2.32%	+ 34.21%	-2.79	+ 0.00832
Q4	37	+ 0.03%	+ 1.11%	+ 59.46%	0.03	- 0.98
Q3	37	-1.39%	-2.77%	+ 40.54%	-1.51	- 0.139
Q2	37	+ 0.03%	+ 1.29%	+ 59.46%	0.04	- 0.972
Q1	38	-0.74%	-0.58%	+ 36.84%	-1.02	- 0.314

symbol_coverage

symbol	hourly_ro...	daily_rows	retained
ALGOUSD	485	21	False
WIFUSD	485	21	False
SUSHIUSD	485	21	False
EGLDUSD	485	21	False
JUPUSD	653	28	False
UNIUSD	653	28	False
TIAUSD	653	28	False
LDOUSD	653	28	False
WIFUSD	653	28	False
FILUSD	653	28	False
FETUSD	653	28	False
XRPUSD	653	28	False
ETCUSD	653	28	False

symbol	hourly_ro...	daily_rows	retained
ARBUSDT	653	28	False
DOTUSDT	653	28	False

Statistics Summary

Key	Value
study	leverage_flow_regime
category	crypto
status	success
rows	194
symbol	BTCUSDT
n_symbols	31
n_retained_symbols	14
n_dropped_symbols	17
n_flow_events	7
corr_flow_btc_fwd	-0.16
latest_flow_z	-0.27
horizon	7
lookback	30
git_commit	ec177e3b52b32af72491c32d027588293e7a328d
executed_at	2026-08-05T08:52:43.005299+00:00

Glossary

Term	Definition
n / count	Number of observations or events in the sample.
edge	Average excess return versus the unconditional baseline.
hit-rate	Share of events with a positive forward return.
p-value	Probability of observing the result under the null of no edge.
*** / ** / *	Significance at 99.9% / 99% / 95% confidence.
α (alpha)	Intercept or active return versus benchmark.
β (beta)	Sensitivity to a reference factor or market.
σ (sigma)	Volatility / standard deviation of returns.

Document Control

Field	Value
Report schema	2
Framework version	0.1.0
Classification	internal
Generated	2026-08-05 08:52 UTC
Author	Cerberus Research
Study slug	leverage_flow_regime
Run ID	20260805_095243
Git commit	ec177e3b
Datasets	duckdb:crypto_binance_taker_ls_ratio, duckdb:crypto_binance_global_ls_ratio, duckdb:crypto_binance_top_ls_positions, duckdb:crypto_fear_greed_index, duckdb:crypto_market_data
Parameters hash	358e611c0453
Parameter · symbol	BTCUSDT
Parameter · benchmark	BTC-USD
Parameter · lookback	30
Parameter · horizon	7
Parameter · z_threshold	2
Parameter · min_obs	200
Parameter · min_daily_obs	60
Data fingerprint	1eebf213accb
Duration	1.1s

List of Figures

- Figure 1: BTCUSDT BTC-USD price, net buy-flow ratio and flow z
- Figure 2: Distribution of 30-day flow z-scores
- Figure 3: BTC 7d forward return vs leverage flow z
- Figure 4: BTC 7d forward return by flow quintile
- Figure 5: Latest net buy-flow z-score by symbol

List of Tables

- Table 1: Daily leverage flow frame tail
- Table 2: Detected buy/sell flow spike events
- Table 3: BTC forward returns by flow quintile
- Table 4: Symbol coverage and latest flow z-scores